STANDARD DRAWING GUIDELINE & CHECKLIST FOR RTU DISTRIBUTION NETWORK				TENAGA NASIONAL	
Doc. No.	ES - RTU-DRWGCL	Rev. No.	3	Date	15 MAR 2022

PREPARED BY				
Name	**Job Title**	**Department**	**Signature**	**Date**
Ir. Ts. Ramlah Binti Awang	Technical Expert (Telecontrol)	Engineering Services		

CHANGE RECORD			
Rev. No.	**Date**	**Responsible Person**	**Description of Change**
0	15 Sept 2017	Ramlah Awang	• Reduce total Tab 11 to Tab 6
1	30 Nov 2020	Ir. Ts. Ramlah Awang	• Edit standard parameter Tab 5 for IEC101 & IEC104 • Add IOA standard
2	15 Feb 2021	Ir. Ts. Ramlah Awang	• Check box & reword • Page referance & remarks • Include all module for SGP
3	15 Mar 2022	Ir. Ts. Ramlah Awang	• Change document title & number • Edit standard parameter Tab 5 for DNP3.0 • IO type ID for master sheet

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A. RTU Drawing Review Guideline

1	**Contract No**	
2	**Main Contractor**	
3	**RTU Supplier**	☐Company: ☐Contact Person:
4	**Project Title**	☐SGP Title: ☐Project Title:
5	**Drawing No**	
6	**Voltage Level**	☐ kV PMU/PPU/SSU/PE/Others
7	**RTU Type**	
8	**RTU Model**	
9	**RTU Power Supply**	☐110VDC / 48VDC / 30VDC / 24VDC / 12VDC
10	**Installation Configuration**	☐ Centralized / Networked / Distributed ☐ New / Expansion
11	**Panel Configuration**	Standing floor / wall mounted/ Indoor/ outdoor
12	**Others**	☐ Front page of each tab to be signed and stamped by Professional Engineer

Attachment required-

A. ☐Bill of quantity- SCADA

B. ☐ Valid Sijil Guna Pakai (SGP)

C. ☐Minute of Meeting specified scope of work (SOW) for SCADA

D. ☐ Other (if any): _____

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TAB 1: TELECONTROL OVERVIEW DRAWING REVIEW

Drawing	Detail Sheet	Respond	Sheet no.	Remarks
RTU Overview Detail on architecture and interface to RTU Note: to mention cable type/core for each interface	Plant connection ☐ RTU-SIP/RCB ☐ RTU-AC-DC ☐ RTU-STNS GND ☐ RTU-COMMS ☐ RTU-GPS	☐ RTU-SIP ☐ RTU-RCB ☐ RTU-AC-DC ☐ RTU-STSN GND ☐ RTU-COMMS ☐ RTU-GPS		Ref. to BQ/SOW
	Node-to-node (for different cubicle) / Slave			
	Others (special connection RTU-Gateway/IEDs)			
	Communication system (Protocol, Medium & connection)			

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TAB 2: RTU CONFIGURATION DRAWING REVIEW

a. RTU card information: (Ref. to BQ/SOW)

Module	No. Of Ports	Total I/O (BQ)	Total I/O (This RTU)	Total card	File/Rack/Slot Number
PSU					
CPU					
Communication					
Digital Input					
Digital Output					
Analog Input					
Analog Output					
Others					

Drawing	Detail Sheet	Respond	Sheet no.	Remarks
General	RTU Model	RTU Model: **CPU FW version:**		Ref to Valid SGP
	RTU software version. Note: as SGP approved list software version	RTU software version:		
	Conformal coating			Ref. to BQ/SOW
	IEC 1331 Programmable Algorithm			
	GPS			
	Others			
	Input Supply : **110/30Vdc/24Vdc**	☐ 110Vdc ☐ 30Vdc ☐ 24Vdc ☐ 12Vdc ☐ others		

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	RTU configuration ☐ Centralized ☐ Decentralized	☐ Centralized ☐ Decentralized		
	RTU Panel Configuration ☐ One ☐ Two	☐ One ☐ Two		
Communication Ports	Communication Protocols ☐ IEC101 ☐ IEC104 ☐ DNP3.0	☐ IEC101 ☐ IEC104 ☐ DNP3.0		
	Communication Port configuration ☐ Channel/Port number ☐ Function ☐ Physical Interface	☐ Channel/Port number ☐ Function ☐ Physical Interface		
Rack/Module Configuration	☐ Rack/Module configuration Note: To include module name, part number, firmware version, I/O configuration (wet or dry contact)			Ref to Valid SGP
	☐ Typical Jumper setting (default & required setting) for each module (CPU and I/O)			
Spare Parts	RTU SPARE PARTS LIST DRAWING REVIEW Note: To list detail module for this contract (module/ part number/ firmware version/ manufacture/ Quantity supplied/ others). To indicate number to be supplied under this contract.			Ref. to BQ/SOW & SGP
Special Tools	RTU SPECIAL TOOLS LIST DRAWING REVIEW Note: To list detail (part number/ version/ manufacture/ Quantity supplied/ others) RTU configurator, RTU diagnostic software and hardware firmware version. To indicate number to be supplied under this contract.			

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TAB 3: RTU CUBICLE INTERNAL WIRING DRAWING REVIEW

Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
DC Distribution Circuit Diagram	DC distribution circuit	Ref. to BQ/SOW: for Single panel and double panel	
AC Distribution Circuit Diagram	AC distribution circuit		
Grounding System	Grounding circuit		
Digital Input Module Circuit Diagram	DI module circuit		
Analog Input Module Circuit Diagram	AI module circuit		
Control Output Module Circuit Diagram	CO module circuit		
Analog Output Module Circuit	AO module circuit		
Control Enable Switch Circuit Diagram *Pre-approved Standard drawing*	Control Enable Switch circuit with Buzzer circuit		
RTU Communication Module Wiring – RCC *Pre-approved Standard drawing*	RCC Communication circuit • IEC101 • IEC104 • DNP3.0	***To submit 2 sheets 1 for modem/LL and another sheet for FO***	
RTU Communication Wiring – IED and Satellite RTU *Pre-approved Standard drawing*	IED Communication circuit • IEC103 • Others	Ref. to BQ/SOW	
RTU Internal Subracks or Inter Module Wiring	RTU subrack wiring		
RTU Dummy CB Circuit *Pre-approved Standard drawing*	RTU Dummy CB circuit		
RTU Fault Alarm Circuit Diagram *Pre-approved Standard drawing*	RTU Fault Alarm circuit		

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TAB 4: RTU CUBICLE DRAWING REVIEW

Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
RTU Modules – Parts List (per contract)	Type of modules Note: to include module/ part number/ firmware version/ manufacture/ specification/ quantity.	The information shall consistent with Tab 2	
Electronic Modules /Accessories – Parts List	DC Surge Arrestor	Ref. to BQ/SOW	
	Communications Surge Arrestor □ Communication module for IEC101/IEC104/DNP3 □ modem 30V / 50V		
	Modem □ Pilot/Lease Line □ Fiber □ others		
	Splitter □ Active □ Passive		
	Dummy CB		
	Buzzer		
	Others		
Panel Accessories – Part List	DC MCB	Ref. to BQ/SOW	
	AC MCB		
	I/O Terminal Block- knife type Note: to provide plastic cover & caution sticker for control circuit		
	AC/DC Terminal Block – sliding type with plastic cover & caution sticker		
	Panel Lamp – non florescent		

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Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
	Communication Terminal Block – sliding type		
RTU Cubicle Specification Sheet *Pre-approved Standard drawing* *Indoor* *Outdoor* *Floor standing* *Wall mounted*	RTU Manufacture:		
	Dimension incuding plinth (100mm): □ Standing Floor: *2300mm(H) x 800mm(W) x 800mm(D)* □ Wall mounted: 700mm-900mm (H) x 600mm-800mm (W) x 300mm-400mm (D)		
	Thickness: □ *2mm side, front, rear and door* □ *3mm top and bottom* □ *3mm top and bottom gland plates*		
	Painting Exterior: *RAL 7032*		
	Degree of protection □ IP51 (indoor) □ IP54 (outdoor)		
	□ IP Certificate: □ Panel manufacture:		
	Material: ***Steel***		
Front View	Overall arrangement (door removed)		
Overall arrangement	□ Rear View □ Left & Right View □ Top and Bottom View		
Others	□ Overall subrack □ module arrangement □ Terminal Block arrangement		
Label List	Complete Label List		
RTU Cubicle Name plate	Label on Name plate Minimum information:		

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Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
Note: *Stainless steel pan head screws*	RTU Type & Model, Serial Number, Contract Number, Manufacture Date, Panel IP rating Note: SAT date to be available in the sticker form		
	Material		
	A retractable shelf or equivalent tray shall be provided in the panel to place RTU maintenance tools and at a proper height.		

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TAB 5: RTU PARAMETER LIST DRAWING REVIEW

<table>
<tr><th>Drawing</th><th>Detail Sheet</th><th>Remarks</th><th>Respond/ Sheet No.</th></tr>
<tr><td rowspan="3">Configurable Parameters

Pre-approved Standard drawing</td><td>RTU Level</td><td></td><td></td></tr>
<tr><td>RCC Protocol Level</td><td></td><td></td></tr>
<tr><td>Module Level</td><td></td><td></td></tr>
<tr><td>Programmable Algorithm

Pre-approved Standard drawing</td><td>List of programmable algorithms</td><td></td><td></td></tr>
<tr><td>Parameter

Pre-approved Standard drawing</td><td>Parameter setting for all protocol master/ sub-master/ main-standby

☐ IEC101
☐ IEC104
☐ DNP3.0

Note: Standard parameter setting as attached.</td><td>Appendix 1: Standard Parameter setting</td><td></td></tr>
<tr><td>RTU Master Sheet</td><td>Master Sheet

Table Standard IOA address

<table>
<tr><th colspan="3">IOA Address (IEC10, IEC104 & DNP3)</th></tr>
<tr><td rowspan="5">Digital Input</td><td>DB- Dummy Breaker (TID 31)</td><td>1</td></tr>
<tr><td>DB (TID 31)</td><td>2-100</td></tr>
<tr><td>SB- CES (TID 30)</td><td>101</td></tr>
<tr><td>SB (TID 30)</td><td>102-949</td></tr>
<tr><td>SB – Virtual Point (TID 30)</td><td>950-1000</td></tr>
<tr><td rowspan="3">Digital Output</td><td>DC- Dummy Breaker (TID 46)</td><td>1001</td></tr>
<tr><td>DC (TID 46)</td><td>1002-1500</td></tr>
<tr><td>SC (TID 45)</td><td>1501-2000</td></tr>
<tr><td rowspan="2">Analog input</td><td>AI ((TID 36)</td><td>2001-2099</td></tr>
<tr><td>AI for RE- Virtual Point (TID 36)</td><td>2100-2103</td></tr>
</table>

Note: Actual Master Sheet shall be approved before FAT. Standard Master sheet as attached.</td><td>Appendix 2: Standard Master sheet</td><td></td></tr>
</table>

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Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
AC, DC and Comms Cable Schedule *Pre-approved Standard drawing*	AC Cable		
	DC Cable		
	Communication Cable		

TAB 6: RTU CABLE SPECIFICATION DRAWING REVIEW

Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
Cable Overview	Cable function/ construction/ no.of core/ conductor size/ conductor/ standard	Ref. to BQ/SOW	
Cable Specification Label on Cross Sectional Diagram *Pre-approved Standard drawing*	**Digital Input cable** ☐ 600/1000V PVC/PVC/SWA/PVC 1.0 mm sq, Multi-core Auxiliary Cables (MS274 or BS 6346)	Shall comply to: TECHNICAL SPECIFICATIONS ***TELECONTROL SYSTEM CABLES & CABLING*** ***KEJ10403:2015***	
	Digital Output Cable ☐ 600/1000V PVC/PVC/SWA/PVC 1.0 mm sq Multi-core Auxiliary Cables (MS274 or BS 6346)		
	Analog Input cable ☐ 300/500V PVC/PVC/SWA/PVC 1.0 mm sq, Multi-pair Twisted, Screened (BS5308)		
	Communication cable ☐ 300/500V PVC/PVC/SWA/PVC 1.5 mm sq, 5-pair twisted, overall screened, Instrumentation Cables (BS 5308)		

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Drawing	Detail Sheet	Remarks	Respond/ Sheet No.
	Communication Fiber Optic ▯ Armoured Single Mode Fibre Optic Cable with minimum 2 cores per bundle (IEC 60794).		
	Ground cable ▯ PVC 50mm2. Power Cable Specification: BS 6004 ▯ Conductor shall be plain annealed high conductivity copper wire with a minimum size of 50.0 mm sq (stranded) to BS 6360		
	DC cable ▯ PVC/PVC/SWA/PVC min 4.0mm2, stranded. Power Cable, 600/1000V, Specification: BS 6346		
	AC cable ▯ PVC/PVC/SWA/PVC min 4.0mm2, stranded Power Cable, 600/1000V, Specification: BS 6346		

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APPENDIX 1

PARAMETER/CONFIGURATION CHECK FOR IEC 101/IEC104/DNP3

Parameters/ Configurations	Detail Settings	Description	This RTU
Baud Rate	9600 □ Unbalance IEC 101, □ Balance DNP3/IEC 104		
Start Bit	1		
Stop Bit	1		
Data Bit	8		
Parity	Even / None		
Buffer Size	1000 (delete newest)		
Communication Time Out	25 sec / 30 sec / 60 sec		
Waiting for Next Poll	20 sec		
Time Sync	Master station • 60 min		
Module Level:	Digital Input &Analog Input • Class 2		
Channel Redundancy	Main-Standby		
Switch channel (main-standby)	20 Sec / 25 sec		
Summer Time	Disable		
GPS time	Disable		
Deadband	50 Count / 100 Count / % of Changes		
Debounce	10 msec		
Chatter, Nc	5 Count		
Chatter Time, Tc	1 sec		
Recovery Time	1 min		
Double Bit Processing	10 sec		

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Parameters/ Configurations	Detail Settings	Description	This RTU
DO Contact Closure ms	2.5 sec / 3 sec		
Control Time Out	20 sec		
IEC 104			
TCP port no.	2404		
t0	30 sec		
t1	15 sec		
t2	10 sec		
t3	20 sec		
K	12		
W	8		
DNP3.0			
DI	☐ SB Class 1, G2 V2 ☐ DB Class 1, G4 V2		
AI	☐ Class 2		
DO	☐ Class 2, G12 V1	**CROB**	

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APPENDIX 2

SAMPLE STANDARD MASTER SHEET FOR RTU DISTRIBUTION NETWORT

RACK / SUB-RACK	DESCRIPTION	SIGNAL TYPE	MODULE	IEC 101 IOA	IEC 104 IOA	Type ID	Remarks
	DOUBLE BIT						
	DUMMY SWITCH OPEN	DB	DI	1	1		1-100 for DB
	DUMMY SWITCH CLOSE						
	CB 1 OPEN INDICATION	DB	DI	2	2		
	CB 1 CLOSE INDICATION						
	CB 2 OPEN INDICATION	DB	DI	3	3		
	CB 2 CLOSE INDICATION						
	.			4	4		
	.						
	DS MAIN OPEN INDICATION	DB	DI	20	20		DS = Disconnector
	DS MAIN CLOSE INDICATION						
	DS RESERVE OPEN INDICATION	DB	DI	21	21		
	DS RESERVE CLOSE INDICATION						
	EARTH SWITCH OPEN INDICATION	DB	DI	25	25		
	EARTH SWITCH CLOSE INDICATION						
	.			.	.		
		DB	DI	100	100		
	SINGLE BIT						
	RTU CONTROL ENABLE SWITCH	SB	DI	101	101		101-949 for Single Bit
	CB 1 REMOTE / SUPERVISORY SWITCH	SB	DI	102	102		
	CB 1 OVERCURRENT RELAY INDICATION	SB	DI	103	103		
	CB 1 EARTHFAULT RELAY INDICATION	SB	DI	104	104		
	.	SB	DI	.	.		
	.	SB	DI	.	.		
	DIGITAL INPUT SUPPLY FAIL (example)	SB	DI	948	948		RTU specific alarm
	DIGITAL OUTPUT SUPPLY FAIL (example)	SB	DI	949	949		RTU specific alarm

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RACK / SUB-RACK	DESCRIPTION	SIGNAL TYPE	MODULE	IEC 101 IOA	IEC 104 IOA	Type ID	Remarks
VIRTUAL POINT							
	RTU BUFFER FULL ALARM	VP	DI	950	950		**950-1000 for Virtual Point**
	RTU TIME SYNC ALARM	VP	DI	951	951		
	RTU HEALTY ALARM	VP	DI	952	952		
	RTU COMMUNICATION TO NCC FAIL	VP	DI	953	953		
	DIGITAL INPUT FAIL (DI 1-n)	VP	DI	954	954		
	DIGITAL OUTPUT FAIL (DO 1-n)	VP	DI	*	*		Running number after DI module(s)
	ANALOG INPUT FAIL (AI 1-n)	VP	DI	*	*		Running number after DO module(s)
	RE GAETWAY/IED FAIL	VP	DI	*	*		
ANALOG INPUT							
	CB 1 CURRENT (4-20mA) ((example))	AI	AI	2001	2001		**2001-2999 for AI**
	CB 2 CURRENT (4-20mA) ((example))	AI	AI	2002	2002		
	.			.	.		
	.			.	.		
	CB 3 VOLTAGE (4-20mA)	AI	AI	2007	2007		
	CB 4 MWATT (4-12-20mA)	AI	AI	2008	2008		
	CB 4 MVAR (4-12-20mA)	AI	AI	2011	2011		
	RE Voltage (protocol)	VP	AI	2100	2100		RE Analog input
	RE Current (protocol)	VP	AI	2101	2101		RE Analog input
	RE P Watt (protocol)	VP	AI	2102	2102		RE Analog input
	RE Q Var (protocol)	VP	AI	2103	2103		RE Analog input
DIGITAL OUTPUT							
	DUMMY OPEN CONTROL	DC	DO	1001	1001		**1001-1500 for Double Command**
	DUMMY CLOSE CONTROL						
	CB 1 OPEN CONTROL	DC	DO	1002	1002		
	CB 2 CLOSE CONTROL						
	CB 1 OPEN CONTROL	DC	DO	1003	1003		
	CB 2 CLOSE CONTROL						

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RACK / SUB-RACK	DESCRIPTION	SIGNAL TYPE	MODULE	IEC 101 IOA	IEC 104 IOA	Type ID	Remarks
	DS MAIN OPEN CONTROL	DC	DO	1020	1020		DS = DISCONNECTOR
	DS MAIN CLOSE CONTROL						
	DS RESERVE OPEN CONTROL	DC	DO	1021	1021		
	DS RESERVE CLOSE CONTROL						
	MASTER TRIP RESET	SC	DO	1501	1501		**1501-1550 for Single Command**
	EFI RESET	SC	DO	1505	1505		

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